

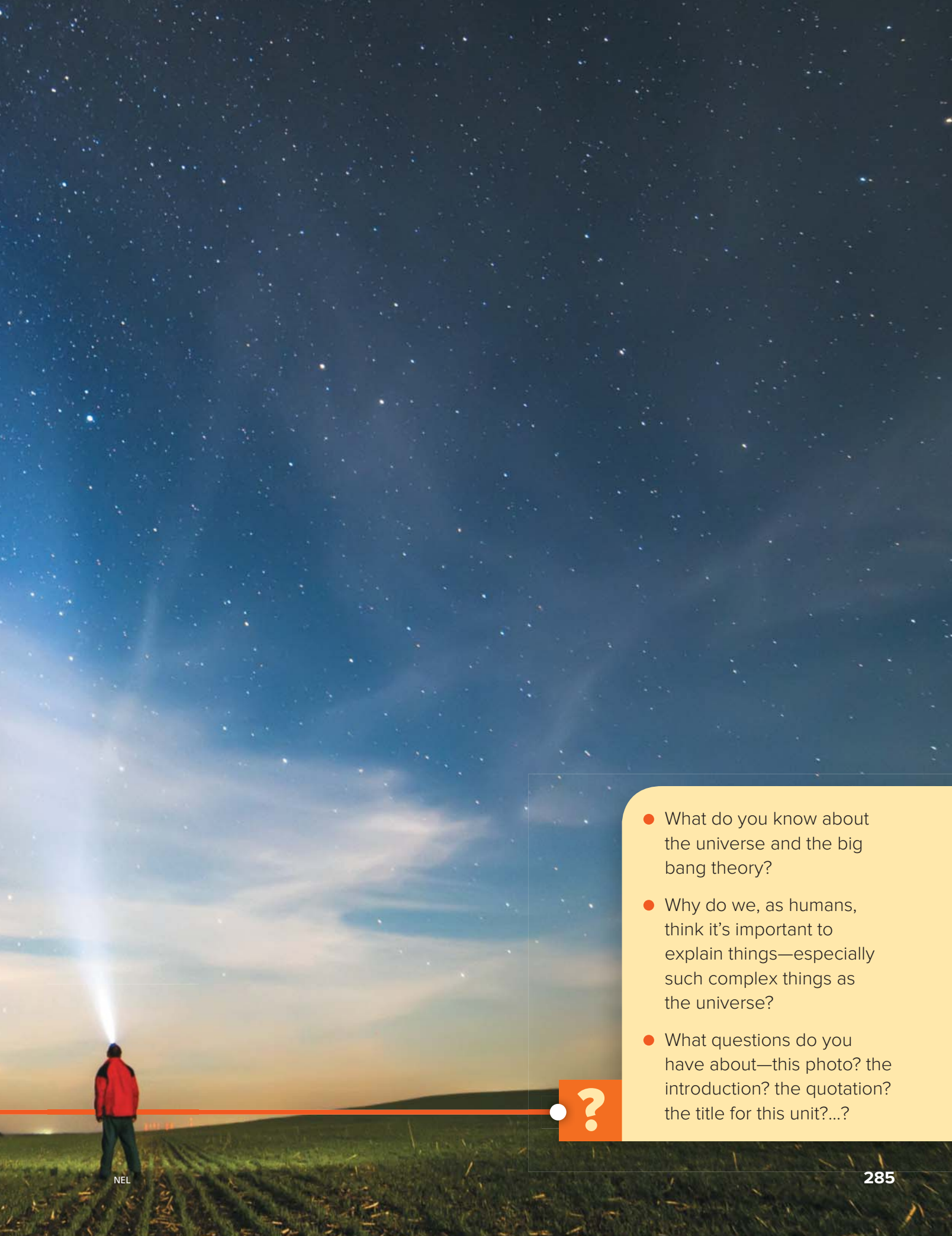
UNIT 4

The formation of the universe can be explained by the big bang theory.

Have you ever shone a flashlight into the vastness of a twilight or night sky? Many people have done so, especially in their youth. Perhaps they wondered if someone might, somehow, from somewhere, glimpse the dim beam. Maybe it helped them feel more connected to a universe that might otherwise seem impersonal. Could it have been a way of calling out to say “Hey, I’m here. I see you. Do you see me?”

“There’s as many atoms in a single molecule of your DNA as there are stars in the typical galaxy. We are, each of us, a little universe.”

*Neil deGrasse Tyson,
astrophysicist, cosmologist,
science communicator*



- What do you know about the universe and the big bang theory?
- Why do we, as humans, think it's important to explain things—especially such complex things as the universe?
- What questions do you have about—this photo? the introduction? the quotation? the title for this unit?...?



At a Glance

You will demonstrate what you know, can do, and understand by being able to

- Perform investigations and use other investigative methods to explore the universe and our understanding of it
- Apply scientific understandings, First Peoples perspectives, and other ways of knowing to analyze, evaluate, and communicate ideas and information about the universe
- Develop and use models and other methods to describe and illustrate objects, events, and processes of the universe

TOPIC 4.1:

What is the universe, and how do we make sense of it?

Some things you will do:

- observe and ask questions about the world around you
- contribute to care for self, others, community, and world through individual or collaborative approaches

Some things you will come to know:

- “Universe” is a word whose meaning depends on various factors.
- Different peoples in different places and at different times in history have different ways to perceive the universe.

ESSENTIAL QUESTION

How can we understand and explain the universe, its formation, and our place in it?





TOPIC 4.2:

What do we know about the universe based on what we can see only with our eyes?

Some things you will do:

- use star maps to recognize different constellations and objects in the night sky
- generate and introduce new or refined ideas

Some things you will come to know:

- In terms of what we can see with our eyes alone, the universe is surprisingly small.
- We can navigate the night sky in ways that are similar to navigating on land.



TOPIC 4.3:

How has technology expanded our knowledge and understanding of the universe?

Some things you will do:

- seek and analyze patterns, trends, and connections in data
- consider changes in knowledge over time as tools and technologies have developed

Some things you will come to know:

- We use technology to help us observe the universe at all wavelengths and frequencies of radiant energy.
- Our ability to catalogue and classify stars has helped us develop ideas about their life cycles.



TOPIC 4.4:

How do we use the big bang theory to explain what we know about the universe?

Some things you will do:

- communicate scientific ideas, claims, and information
- generate and introduce new or refined ideas

Some things you will come to know:

- We have several powerful sources of evidence to support the big bang theory.
- Despite our current state of knowledge, there is still so much for us to learn about the universe.

Connect To What You Already Know

This feature helps you reflect on what you know about some foundational ideas that you have learned in previous grades. Work alone or quietly in small groups to answer the questions. Reach out to your classmates to ask about things that you are unsure about or to offer assistance. Your teacher also can provide additional reinforcement materials to help you prepare for this unit.

1. Each of the following space-related technologies is shown in **Figure 1**. Match each technology to its illustration, and describe how it has contributed to our understanding of space.
 - a) Discovery space shuttle
 - b) Atlas V rocket
 - c) Hubble Space Telescope (HST)
 - d) International Space Station (ISS)

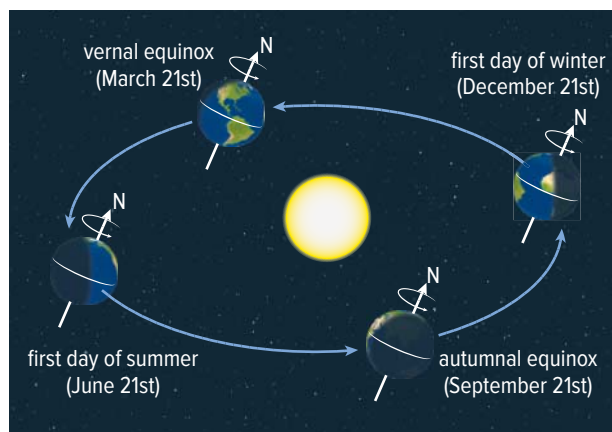


Figure 2 Use this diagram to answer question 4 below.

3. Use a sketch to identify the components of our solar system and their arrangement.
4. It takes 11.86 Earth-years for Jupiter to orbit the Sun. Each Jupiter day is 9 h, 50 min, and 30 s long.
 - a) Which of the above statements describes Jupiter's period of rotation, and which describes Jupiter's period of revolution?
 - b) Distinguish between these motions.
5. Refer to **Figure 2** at the top of this page. Use your understanding of rotation and revolution to explain what is being shown. Your explanation should include the words day, night, seasons, axis, hemisphere, and any other words you think are important.

6. Identify the effect that results from each of the following causes. The causes are listed below. Explain your answer in each case.

- seasons
- Earth's orbit
- yearly periods
- ocean tides
- day and night

a) Earth's rotation on its axis causes...

b) Earth's revolution around the Sun causes...

c) The Moon's gravitational pull on Earth causes...

d) Earth's tilt on its axis causes...

e) The Sun's gravitational pull on Earth causes...

7. Like all scientists, astronomers commonly use scientific notation to express sizes and distances. For example, the diameter of Earth's Moon is 3475 km, or 3.475×10^3 km.

a) Earth's diameter is 12 756 km. Express this measurement in scientific notation.

b) How much larger is the diameter of Earth compared to the diameter of the Moon?

c) After the Sun, the next-nearest star to Earth is Proxima Centauri, at a distance of 40 208 000 000 000 km. Express this distance in scientific notation.

8. The changing appearance of the Moon over a period of about 29 days is referred to as the phases of the Moon, or lunar phases.

a) On paper or using found objects, design a model to show the relationship of the Earth, Moon, and Sun to create lunar phases.

b) When we look at the Moon from Earth, we always see the same lunar features. We never see what is often called "the dark side of the Moon." What motion or combination of motions could explain this?

9. While risky and expensive, space exploration technology is beneficial. For example, the technology that was developed for use in the space shuttles is also used to make better artificial hearts. Express your opinion about the money spent by our federal government for space exploration. In developing your answer, consider how spending on space exploration compares with spending in other areas such as health care, defence, and environmental initiatives. Think about other costs and benefits from space-related spending to help you develop an opinion that is shaped, as much as possible, by evidence.



Unit 4 Preparation *(continued)*

Table 1 Selected Data about Planets of Our Solar System

	Mercury	Venus	Earth	Mars	Jupiter	Saturn	Uranus	Neptune
Average distance from Sun (million km)	57.9	108.2	149.6	227.0	778.6	1433.5	2872.5	4495.1
Distance from Sun (Earth = 1)	0.387	0.723	1	1.52	5.20	9.58	19.20	30.05
Diameter at equator (Earth = 1)	0.383	0.949	1	0.532	11.21	9.45	4.01	3.88
Mass (Earth = 1)	0.0553	0.815	1	0.107	317.8	95.2	14.5	171
Density (Water = 1)	0.984	0.951	1	0.713	0.240	0.125	0.230	0.297
Ring System	no	no	no	no	yes	yes	yes	yes
Surface gravity (Earth = 1)	0.378	0.907	1	0.377	2.36	0.916	0.889	1.12
Time for one rotation on axis	58.8	−244	1	1.03	0.415	0.445	−0.720	0.673
Time for one revolution of Sun	0.241	0.615	1	1.88	11.9	29.4	83.7	163.7
Number of moons	0	0	1	2	67	62	27	14

10. On a large sheet of paper, draw a line 40 cm long. Mark the position to represent the Sun at one end of the line. Using **Table 1**, plot the positions of the planets according to distance.
11. There appear to be two groupings of planets according to distance. Make a list with two columns, one for each grouping. Fill in the planets that belong to each grouping, as well as your reasons for including them.
12. On a second sheet of paper, use a mathematical compass to draw circles that represent the sizes of the planets. Use a scale based on Earth being 1 cm.
13. List the planets most similar to Earth in density.
14. List the planets most similar to Earth in mass.
15. Planets that are similar to Earth are called terrestrial planets or rocky planets. Which planet is Earth-like in size but very different other ways?
16. Identify the terrestrial planets.
17. List the four planets that are usually referred to as the gas giants.
18. Saturn, with a diameter that is almost 9.5 times that of Earth, has a surface gravity fairly close to that of Earth. (Gravity is the force of attraction that acts between any given pair of objects.) Use the table to determine how two planets with great differences in size could have almost the same surface gravity.

19. Uranus and Jupiter have nearly identical densities, but Jupiter is much larger. Which would you expect to have a larger force of gravity? Explain why, and then check your prediction in the table.
20. Which planet has the shortest year? Which has the longest year?
21.
 - a) Make a statement that relates the distance of a planet from the Sun and the length of its year.
 - b) Does this statement apply in terms of the length of a planet's day? Explain.
22. Most planets have at least one moon. Why do you think Mercury and Venus have none?
23. What other data and facts do you know or can you find out about the planets of our solar system? (For example, what do those negative signs for the rotation of Venus and Uranus indicate?)
24. Our solar system has two regions that are referred to as "belts." Add these regions to the diagram you made in question X. What are these regions called, and how do their components compare?
25. What are comets and where are the majority of comets of our solar system thought to originate?
26. Distinguish among the following terms: meteoroid, meteor, meteorite.
27. Brainstorm ways in which stars differ from planets. Organize your comparisons in the form of a T-chart or a Venn diagram.
28. **Figure 3** shows an imaginary solar system far, far from our own. Astronomers have classified the five planets as shown, and have determined their distances from the star they orbit. Astronomers think that there is a sixth planet in this solar system, and it is located somewhere in the region marked X.
 - a) What type of planet would you expect to find in this region?
 - b) Explain your answer to a).

- c) At what distance would you expect the new planet to be found?
- d) Why did you choose this distance?
- e) How is this solar system arrangement different from ours?

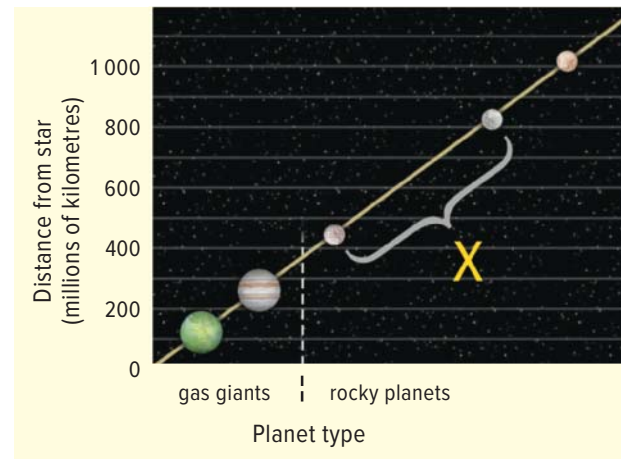


Figure 3 Use this diagram to answer question 28.

29. Describe at least three features of Earth that set it distinctly apart from the other planets of our solar system.
30. Refer to **Figure 4** to answer the following questions.
 - a) What does the diagram show?
 - b) Is it partial or full? How do you know?
 - c) A similar kind of event also involves the Sun, Earth, and Moon. Name this event and make a sketch to show how the Sun, Earth, and Moon are arranged when it is partial and when it is full.

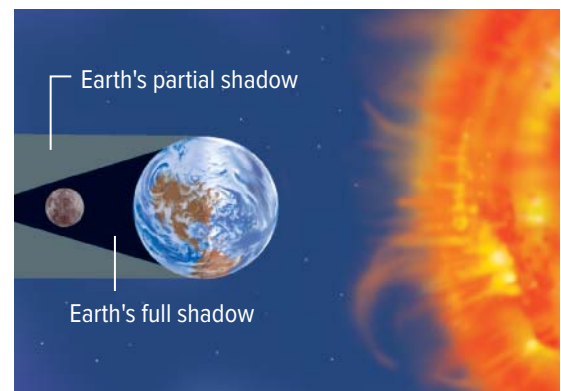


Figure 4 Use this diagram to answer question 29.